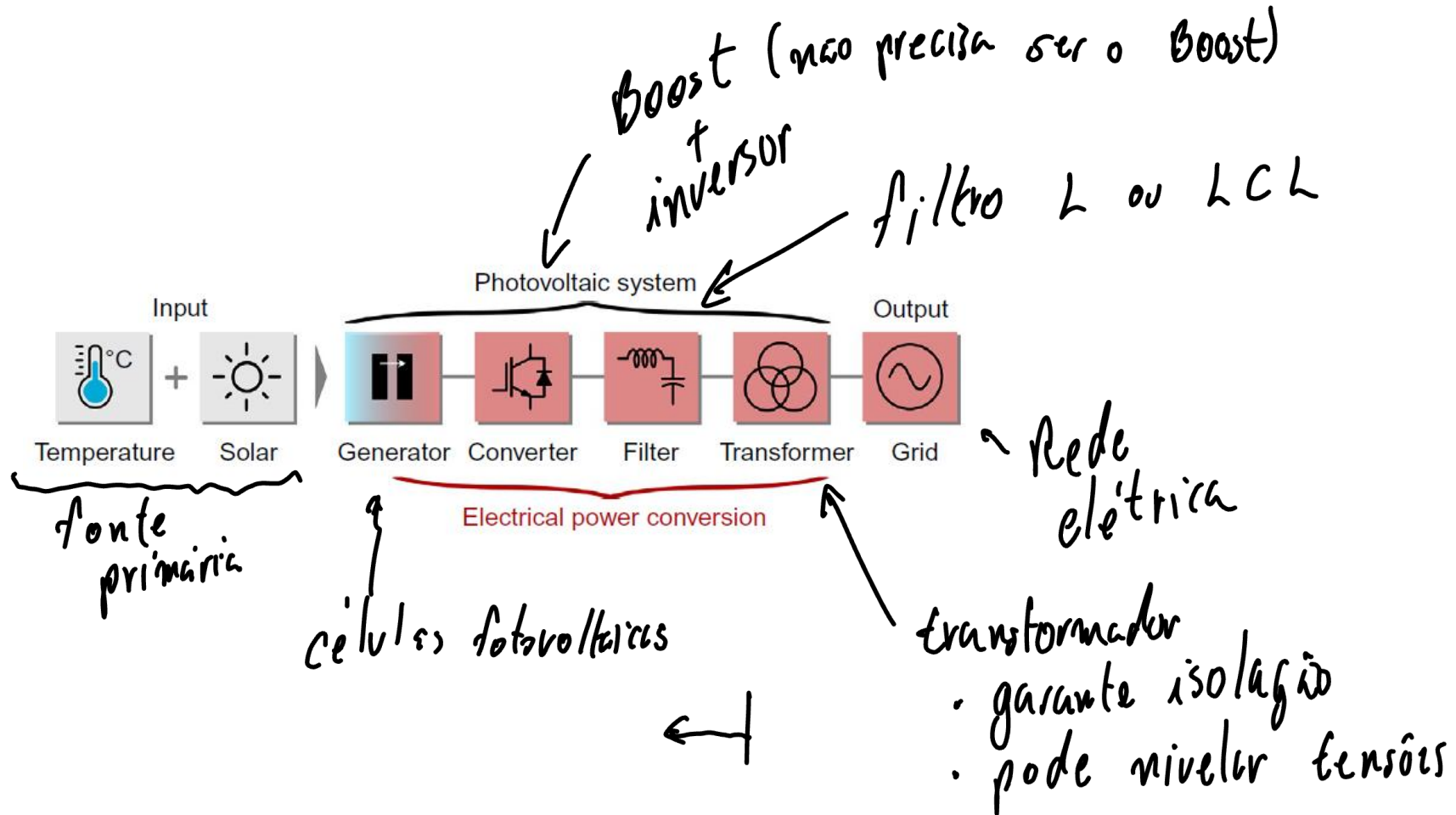
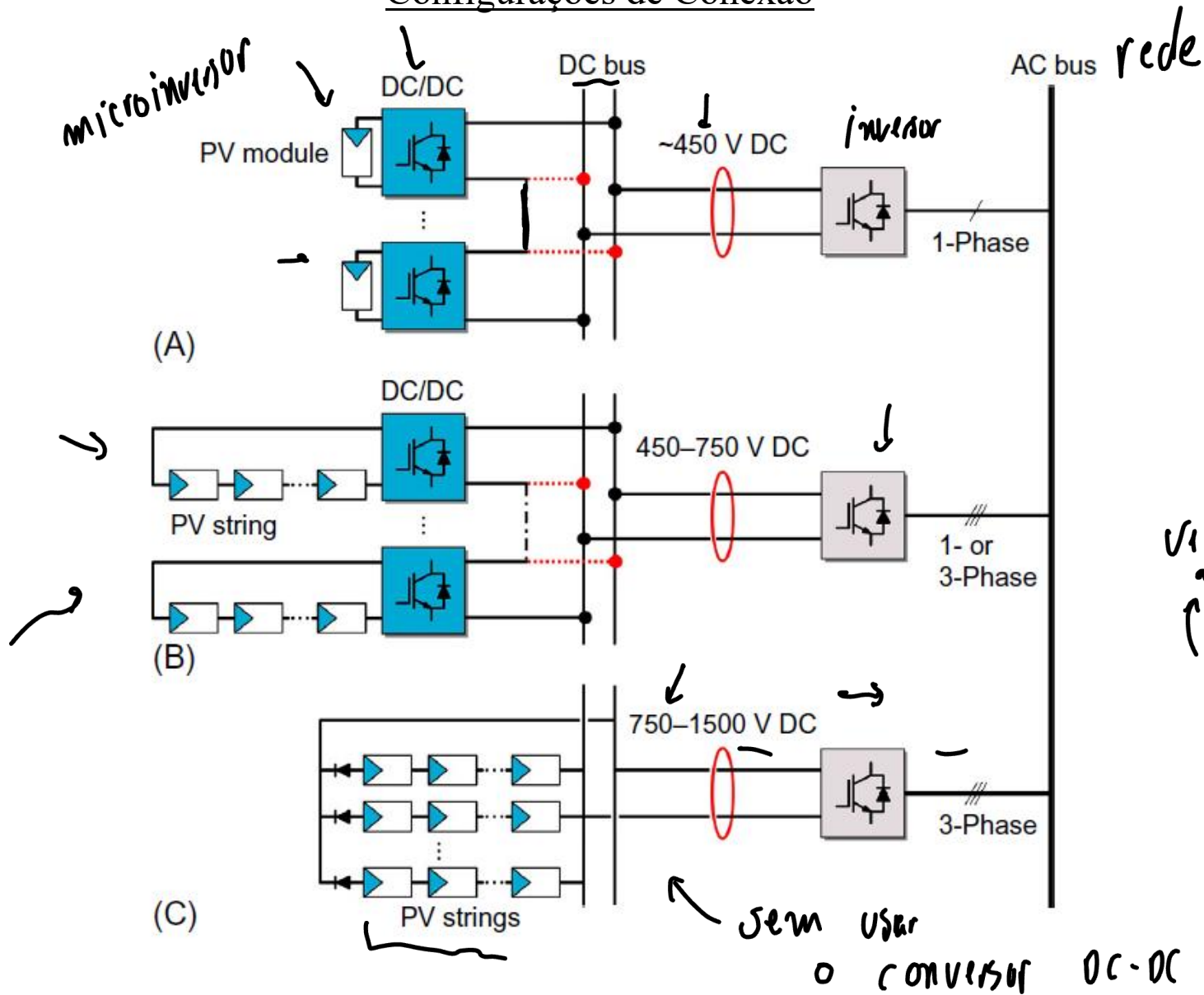


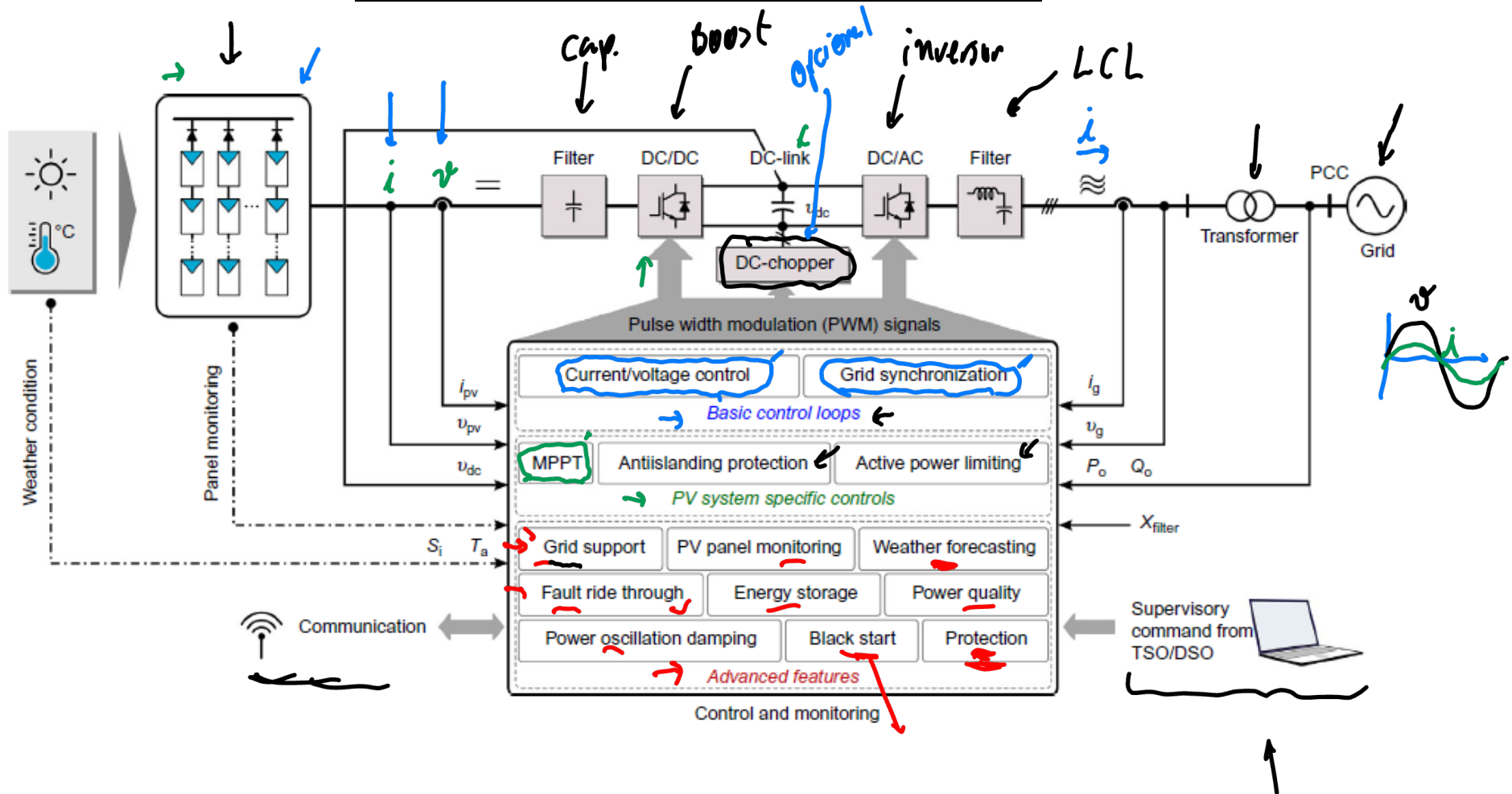
Sistemas Fotovoltaicos conectados à rede elétrica



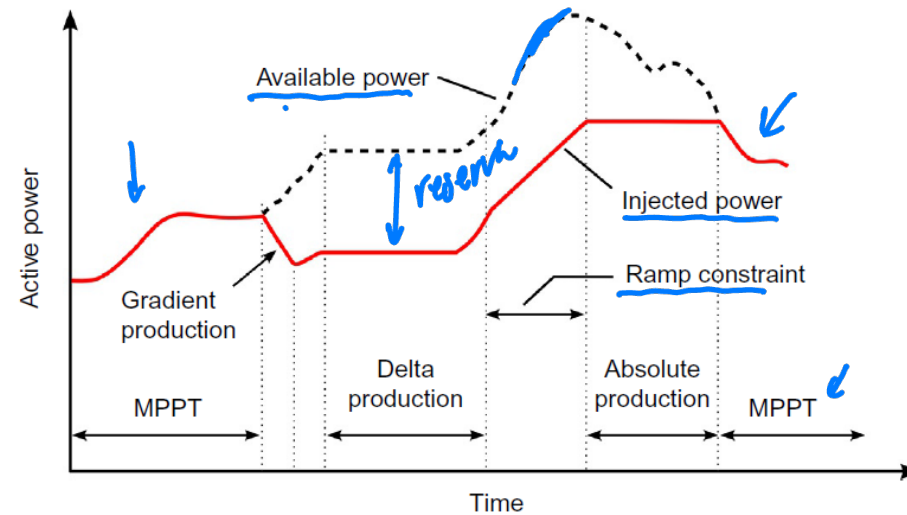
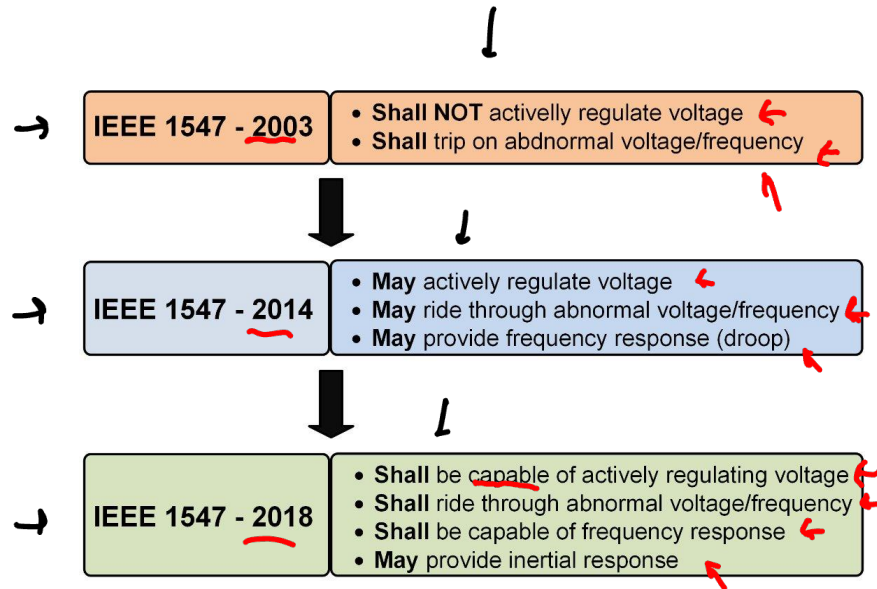
Configurações de Conexão



Visão Geral com conexão DC-DC e DC-AC



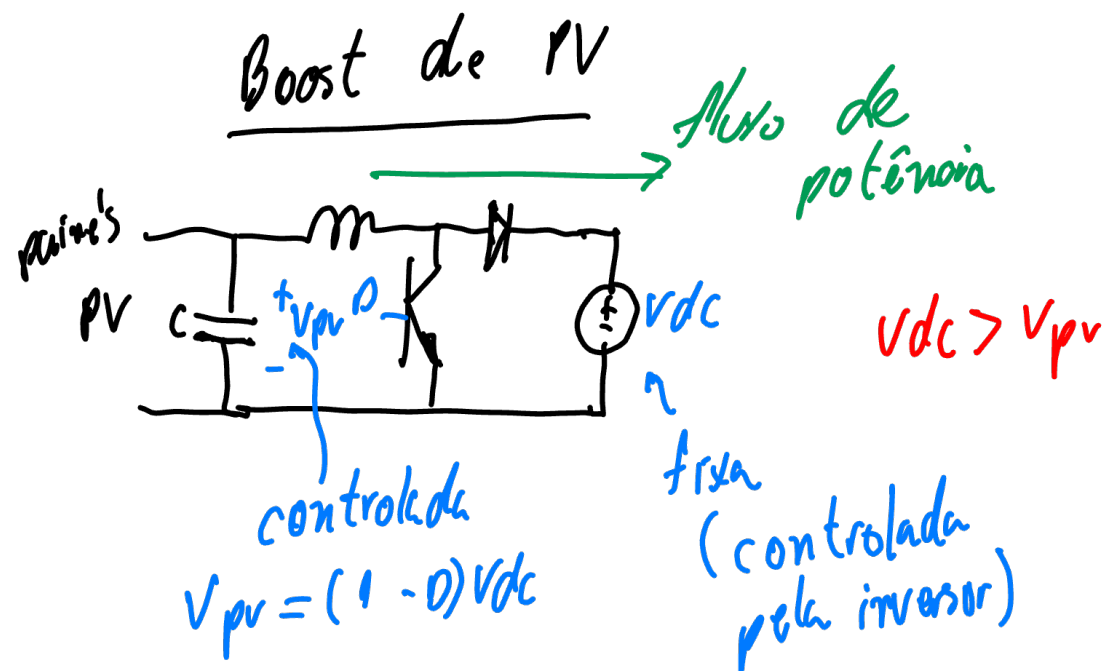
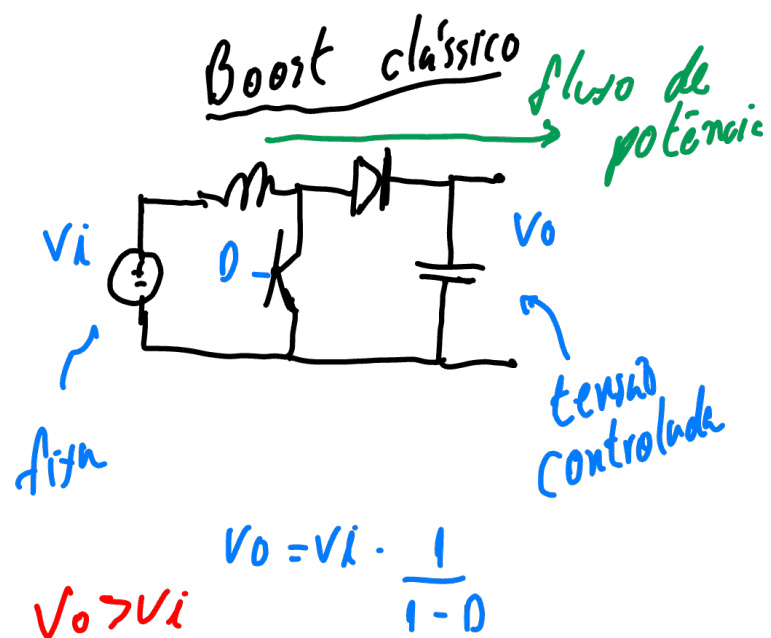
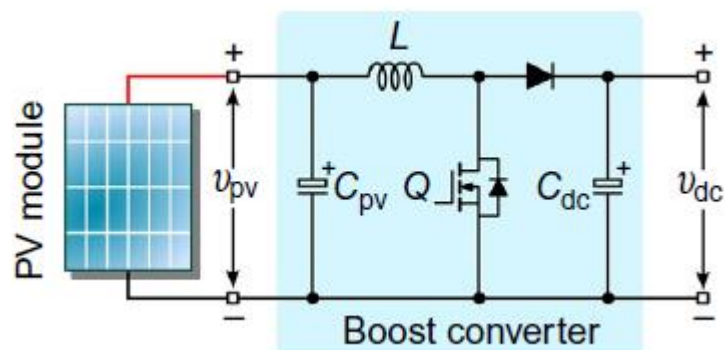
Sobre normas e multifuncionalidade



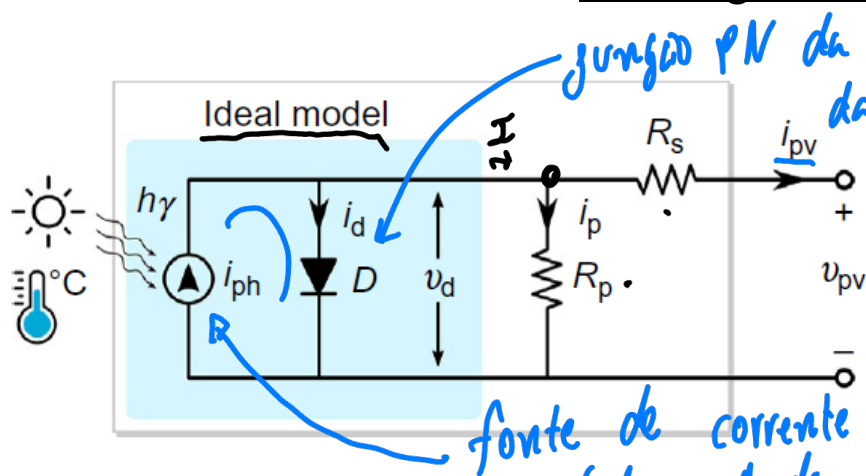
2026

PRODIST - Módulo 3

Considerações iniciais sobre o conversor DC-DC Boost



Modelagem do painel fotovoltaico

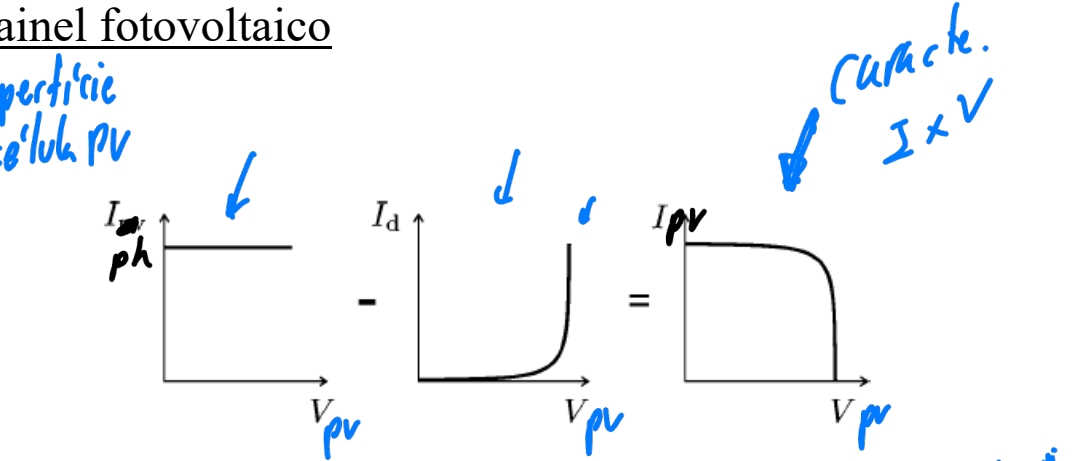


$$i_{pv} = i_{ph} - i_d - i_p$$

↑ ↑ ↑

$$i_d = I_s \left[\exp \left(\frac{v_d}{\alpha V_t} \right) - 1 \right]$$

↑ ↑



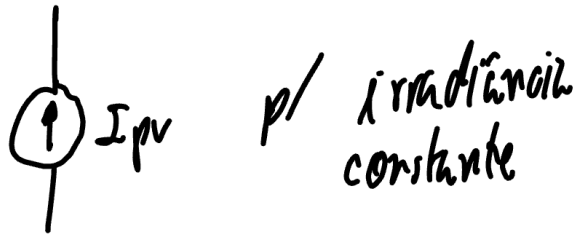
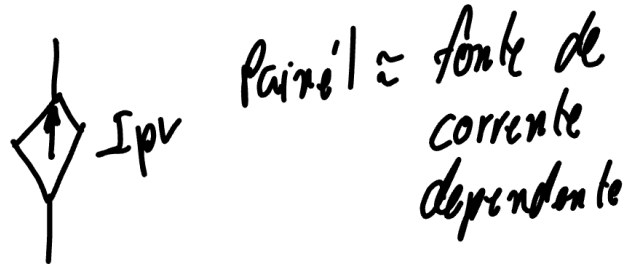
$$i_{ph} = \left[I_{scn} \left(\frac{R_p + R_s}{R_p} \right) + K_i (T - T_n) \right] \cdot \frac{G}{G_n}$$

↑ ↑ ↑

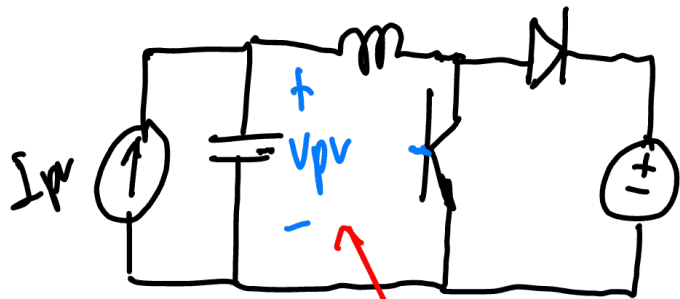
$$i_p = \frac{v_d}{R_p}$$

R_s R_p resistências
parasitas e construção do
painel

Conclusão:



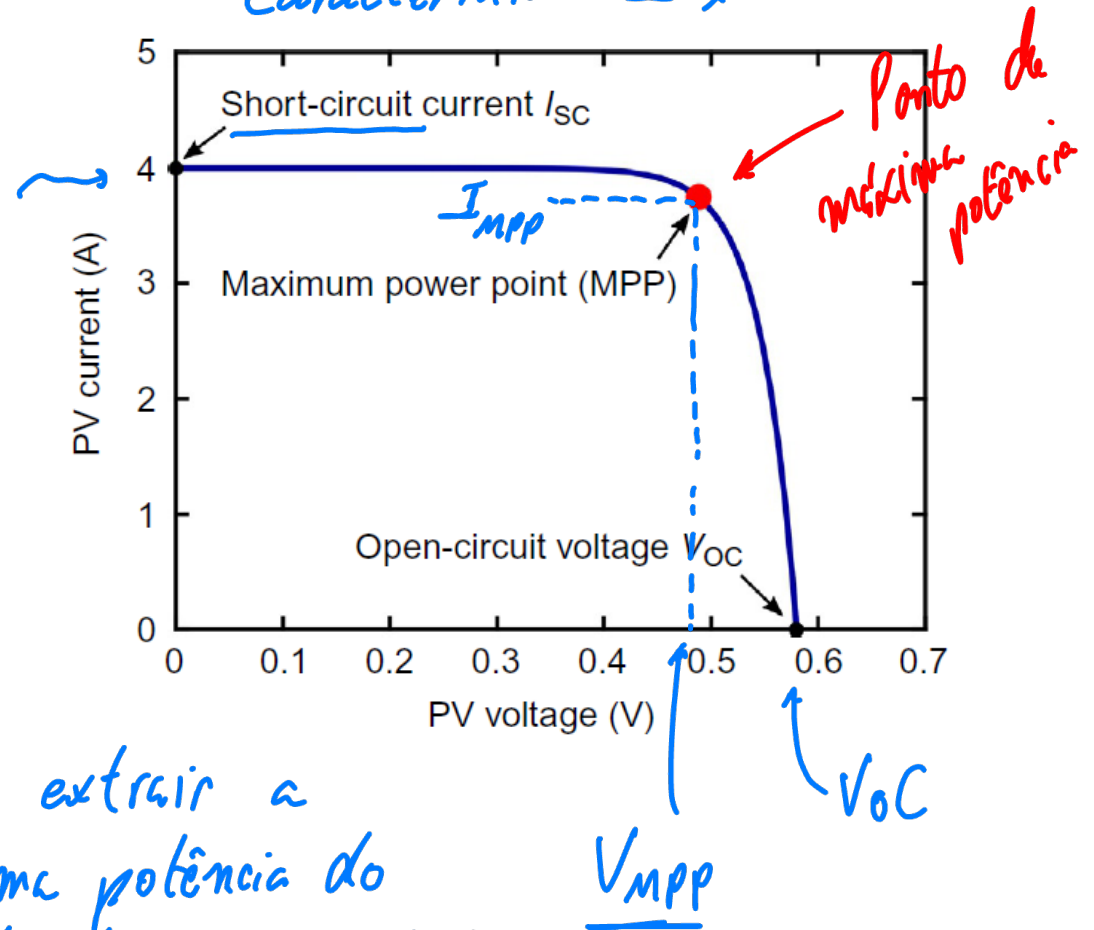
Boost



$V_{pv \text{ ref}}?$

controle em V_{MPP}

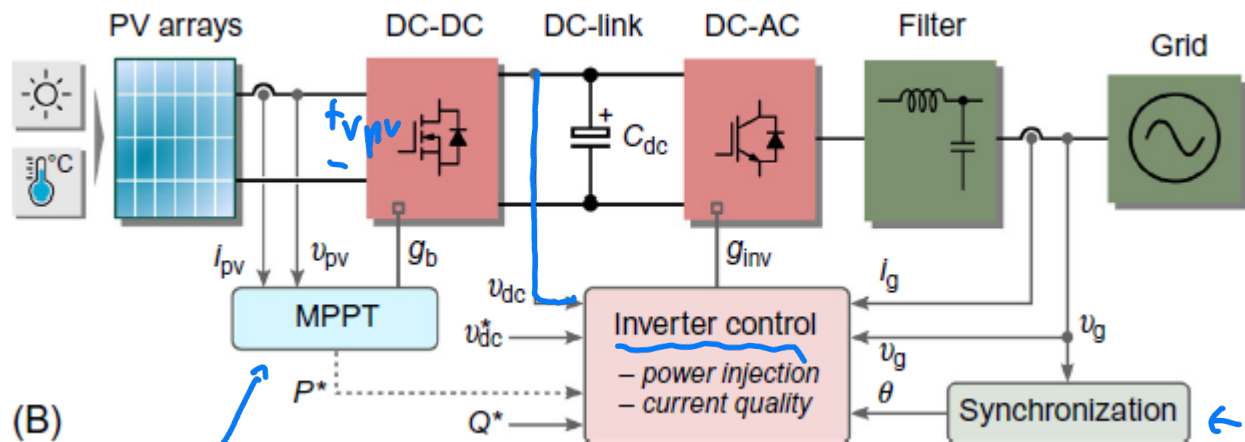
Característica $I \times V$



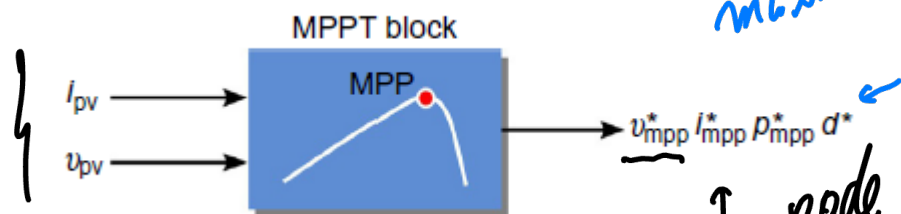
→ Para extrair a máxima potência do painel, devemos controlar a tensão sobre ele em V_{MPP}

→ Como saber qual será o valor V_{MPP} de referência?

Mais sobre a visão geral e algoritmos de rastreamento de máxima extração de potência do painel



↓ algoritmo para
extração da
máxima potência



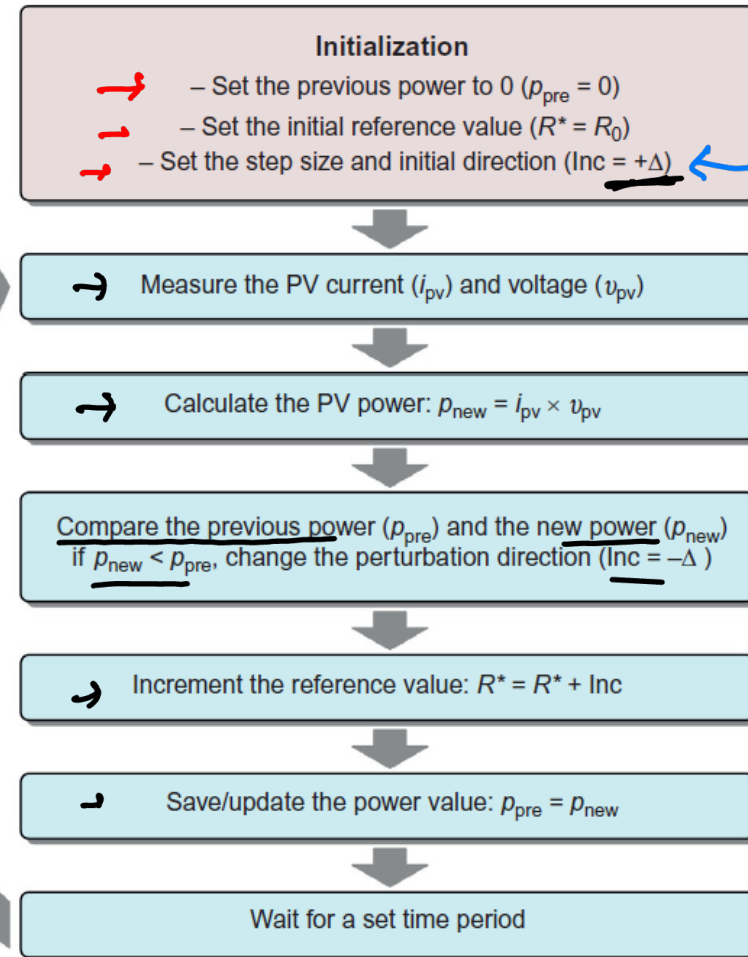
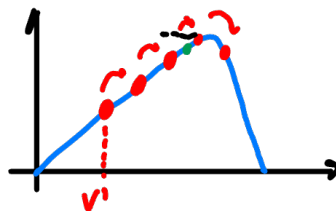
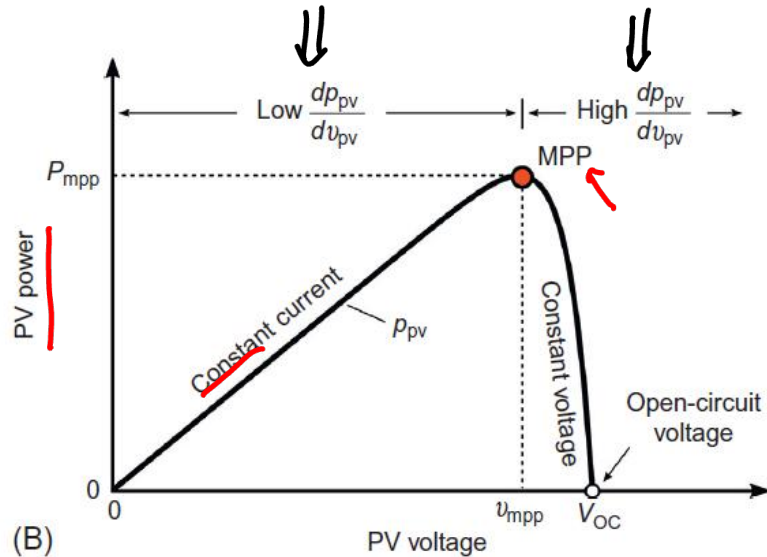
↑ pode gerar
essas referências
→ o mais usual é o v_{mpp}^*

Método Perturba e Observa (P&O)

→ O controle perturba o sistema ao aplicar uma mudança no ponto de operação

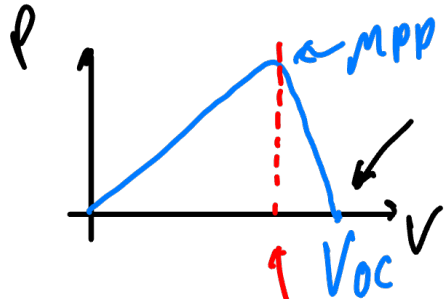
→ Observa a potência e compara com a anterior

→ *



* → Verifica a direção de perturbação
• Se a Potência aumentou, continua perturbando na

Método da Tensão de circuito aberto fracional



*e' uma fracao de Voc
65% a 80%*

$$F_v = 0,8 \text{ (definido)}$$

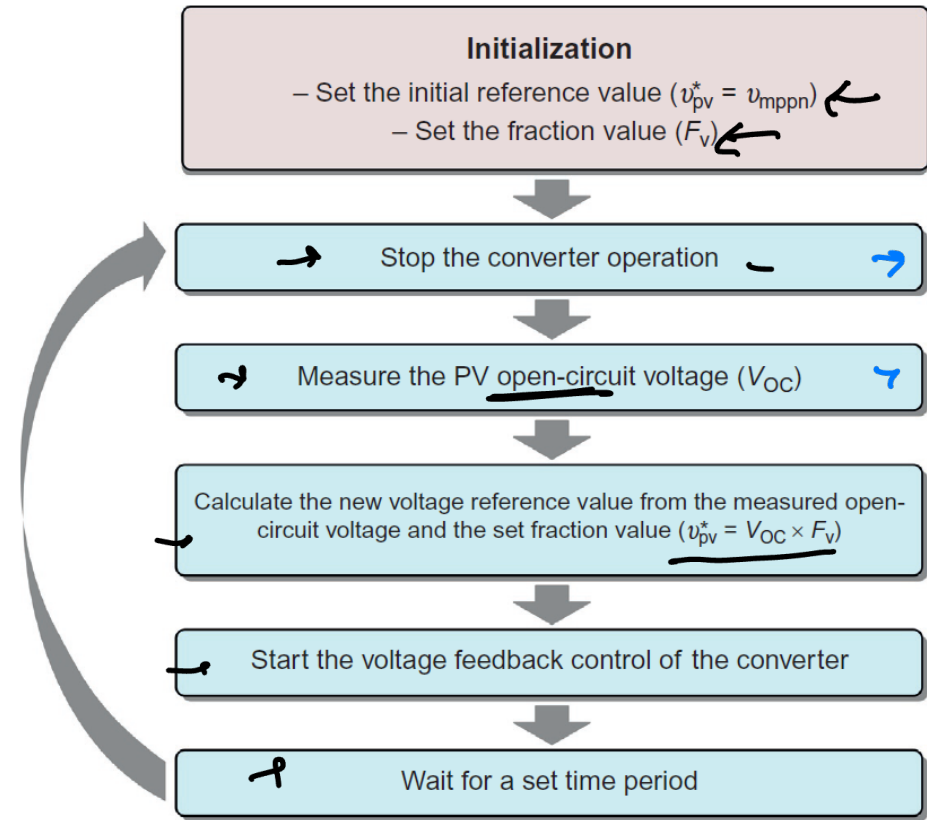
$$V_{pv}^* = F_v \cdot V_{oc}$$

*→ Não garante exatidão
na operação no MPP*

→ só mede o V_{pv}

→ economiza 1 sensor de corrente

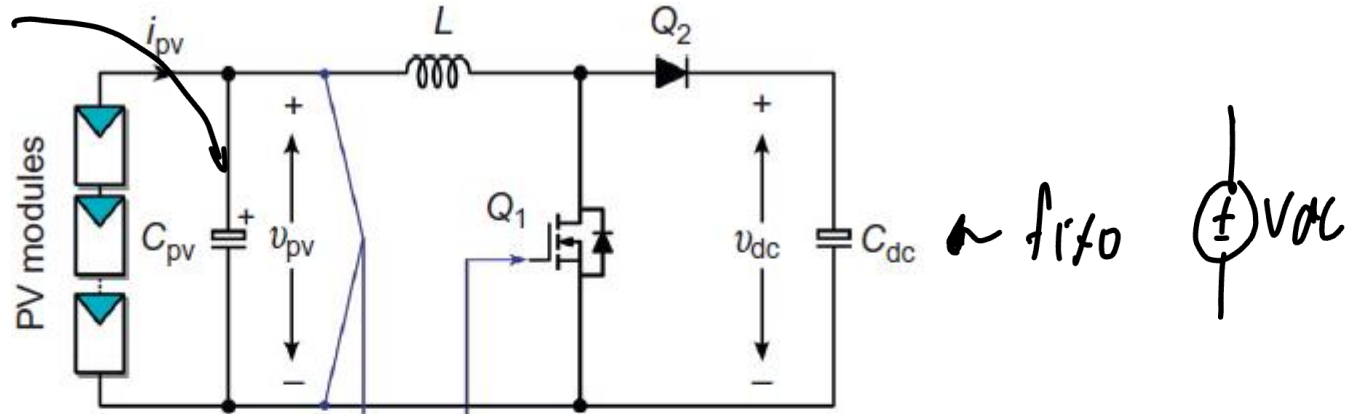
*mesma direção.
• se a potência diminuir,
perturba na direção oposta.*



Para a operação desconecta

Modelagem e controle do Conversor Boost

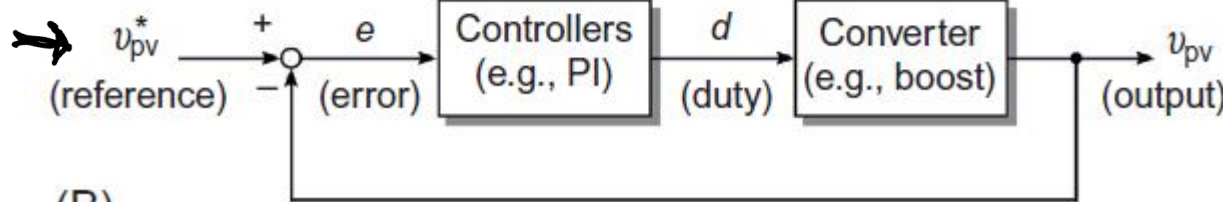
variável
realimentada



Reference
(from MPPT) v_{pv}^*

Feedback
control

(A)

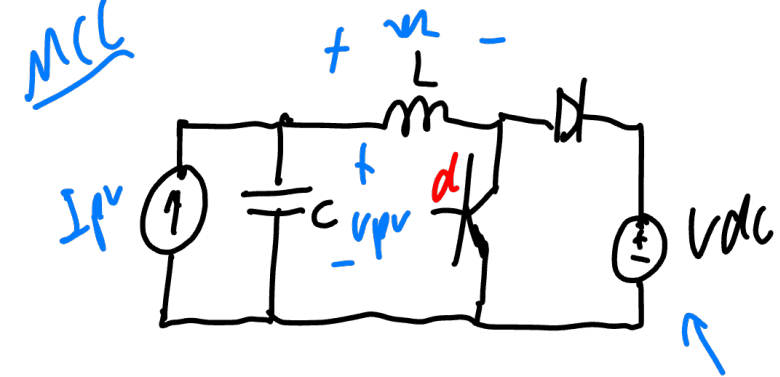


(B)

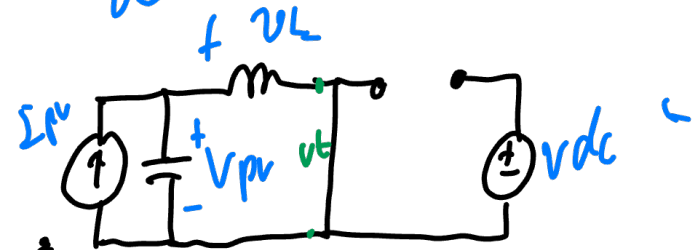
considerado loop interno em
relação ao MPPT. Deve ser mais
rápido que o MPPT. freq. MPPT $\approx$ dezena ou
centena de Hz



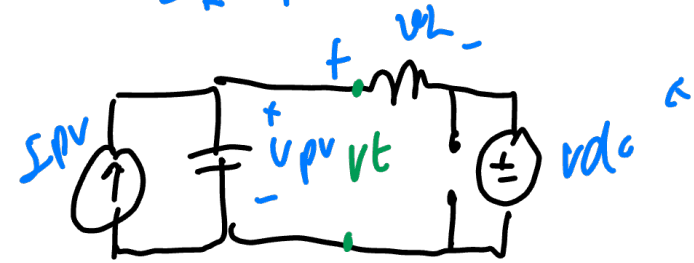
v_{pv}^*
vem
do
algoritmo
de
MPPT



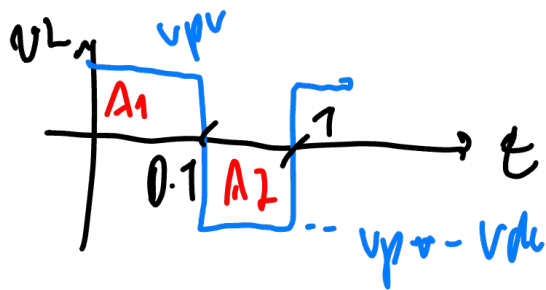
$$\frac{V_{pv}}{d} = ?$$



$$v_L = V_{pv}$$



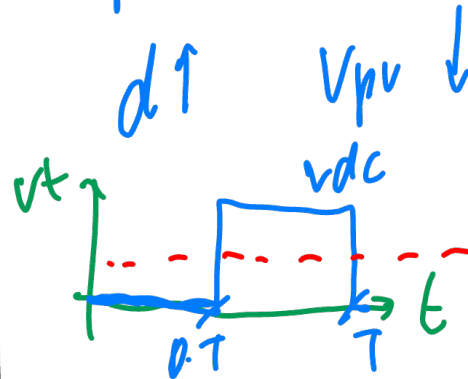
$$v_L = V_{pv} - V_{dc}$$



$$v_{Lmed} = 0$$

$$D \cdot T \cdot V_{pv} = (1-D) \cdot T \cdot (V_{dc} - V_{pv})$$

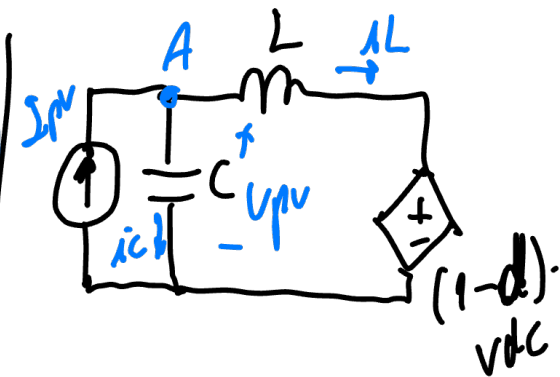
$$V_{pv} = V_{dc} (1-d)$$



$$V_{Cmed} = \frac{A_{vC}}{T}$$

$$= \frac{(1-d) \cdot T \cdot V_{dc}}{T}$$

$$= V_{dc} \cdot (1-d)$$



$$\sum I(A) = 0$$

$$i_{pv} = i_L + i_C$$

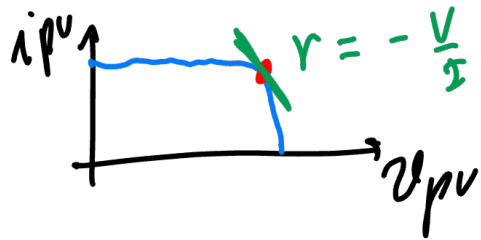
$$i_C = C \frac{dv_{pv}}{dt}$$

$$C \frac{dv_{pv}}{dt} = i_{pv} - i_L$$

$$L \frac{di_L}{dt} = v_L$$

$$L \frac{di_L}{dt} = V_{pv} - (1-d)V_{dc}$$

No ponto de máxima potência,
temos v_{pv} e i_{pv} .



podemos definir uma resistência
incremental

$$r_{pv} = - \frac{v_{pv}}{i_{pv}}$$

$$C \frac{dv_{pv}}{dt} = - \frac{v_{pv}}{r_{pv}} - i_L$$

linearizando

$$v_{pv} = V_{pv} + \tilde{v}_{pv}$$

$$i_L = I_L + \tilde{i}_L$$

$$C \frac{d\tilde{v}_{pv}}{dt} = - \frac{\tilde{v}_{pv}}{r_{pv}} - \tilde{i}_L$$

Laplace

$$C \cdot s \cdot V_{pv}(s) = - \frac{V_{pv}(s)}{r_{pv}} - I_L(s)$$

$$I_L(s) = - \left(C \cdot s + \frac{1}{r_{pv}} \right) V_{pv}(s)$$

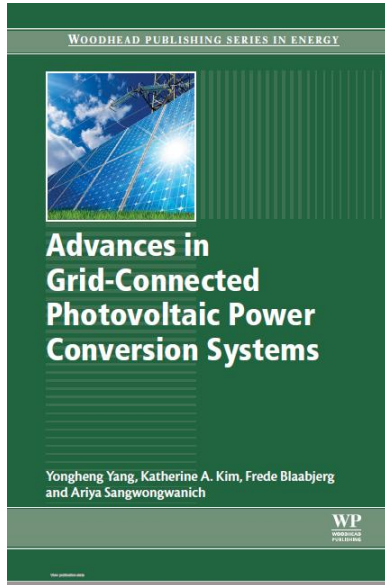
$$L \frac{di_L}{dt} = v_{pv} - (1-d) V_{dc}$$

$$i_L = I_L + \tilde{i}_L$$

$$d = 0 + \tilde{d}$$

$$L \frac{d}{dt} (I_L + \tilde{i}_L) = V_{pv} + \tilde{v}_{pv} - (1 - (0 + \tilde{d})) V_{dc}$$

Referência:



granda sinai: $0 = v_{pv} - (1-d)v_{dc}$

$v_{pv} = (1-d)v_{dc}$

peq. sinai $L \frac{di_L}{dt} = \tilde{v}_{pv} + \tilde{d} v_{dc}$

Laplace

$L s I_L(s) = v_{pv}(s) + D(s) v_{dc}$

$-L \cdot s \left(C s + \frac{1}{r_{pv}} \right) v_{pv}(s) = v_{pv}(s) + D(s) v_{dc}$

O sinal negativo confirma a necessidade de implementar 1-d

$d \uparrow \quad v_{pv} \downarrow$

→ projetar o PI para um freq. de corte pelo menos 10x maior que a freq. MPPT.

$$\frac{v_{pv}(s)}{D(s)} = \frac{-v_{dc}}{L C s^2 + \frac{L}{r_{pv}} s + 1}$$